

Question	Answer	Marks
12(a)	$7\text{}_{-1}^0\text{e}$	A1
12(b)(i)	$E = mc^2$	C1
	$= 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$	M1
	$= 1.494 \times 10^{-10} \text{ J}$ division by 1.60×10^{-13} clear to give 934 MeV	A1
12(b)(ii)	$\Delta m = (82 \times 1.00863\text{u}) + (57 \times 1.00728\text{u}) - 138.955\text{u}$ $= (-) 1.16762 \text{ (u)}$	C1
	energy = 1.16762×934	C1
	energy per nucleon = $(1.16762 \times 934) / 139$ $= 7.85 \text{ MeV}$	A1
12(c)	above $A = 56$, binding energy per nucleon decreases as A increases	B1
	U-235 has larger nucleon number	M1
	so less (binding energy per nucleon)	A1
	or	
	fission takes place with uranium	(B1)
	fission reaction releases energy	(M1)
	binding energy per nucleon less (for uranium than for products)	(A1)